INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Digital Realty PDX11 Data Center, OR -- station KVUO, America/Los_Angeles
Committed pair	OSM 1153935548 -> 734323663 Digital Realty PDX11 Data Center / Digital Realty PDX11 Data Center
Facade gap	66.3 m between the two halls
Weather record	42,062 real hourly records from KVUO
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.8289 C at 270 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-02-28 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 4 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 8 more -- but declared 0 unsafe hours against the agent's 0. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	4 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	7.325	18.0	7.220	5.016
01	mechanical	yes	-	7.537	18.0	7.791	4.658
02	mechanical	yes	-	7.537	18.0	7.791	3.928
03	mechanical	no	refusal	7.446	18.0	7.220	3.335
04	mechanical	yes	-	7.862	18.0	7.220	3.295
05	mechanical	yes	-	8.097	18.0	7.231	3.041
06	mechanical	yes	-	8.087	18.0	7.220	2.309
07	mechanical	no	refusal	9.689	18.0	7.220	2.247
08	FREE	yes	-	11.427	18.0	8.901	2.673
09	FREE	yes	-	12.848	18.0	10.000	3.320
10	FREE	yes	-	15.161	18.0	10.560	4.220
11	FREE	yes	-	17.366	18.0	11.110	5.134

12	mechanical	no	refusal	19.119	18.0	11.670	5.987
13	mechanical	no	refusal	19.105	18.0	12.807	5.706
14	mechanical	no	refusal	20.858	18.0	13.905	5.511
15	mechanical	no	dry-bulb	20.679	18.0	14.156	5.566
16	mechanical	no	dry-bulb	19.767	18.0	12.791	5.338
17	mechanical	no	dry-bulb	18.097	18.0	11.121	4.698
18	mechanical	yes	-	15.877	18.0	9.451	3.782
19	mechanical	yes	-	14.217	18.0	10.571	2.891
20	mechanical	no	refusal	13.638	18.0	8.330	2.859
21	mechanical	yes	switch budget	13.487	18.0	8.330	3.943
22	mechanical	yes	switch budget	12.298	18.0	7.780	4.475
23	mechanical	yes	switch budget	10.670	18.0	6.670	4.712

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.325 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.537 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.537 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.862 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.097 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.087 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.427 C, which is 6.573 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.673 C, and the margin is measured, not chosen: 2.367 C of group-conditional forecast error for this hour of day, plus 0.3064 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.848 C, which is 5.152 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.320 C, and the margin is measured, not chosen: 3.242 C of group-conditional forecast error for this hour of day, plus 0.0782 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.161 C, which is 2.839 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.220 C, and the margin is measured, not chosen: 4.059 C of group-conditional forecast error for this hour of day, plus 0.1606 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.365 C, which is 0.635 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.134 C, and the margin is measured, not chosen: 4.988 C of group-conditional forecast error for this hour of day, plus 0.1455 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.679 C against a 18.0 C limit, so it fails by 2.679 C. It would take a limit 2.679 C higher, or a bound 2.679 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.767 C against a 18.0 C limit, so it fails by 1.767 C. It would take a limit 1.767 C higher, or a bound 1.767 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.097 C against a 18.0 C limit, so it fails by 0.097 C. It would take a limit 0.097 C higher, or a bound 0.097 C tighter, to change this hour.

18:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.877 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.217 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.487 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.298 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.670 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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